

Kevin Selwyn

Software Engineer

(Full resume at <https://kevinsewyn.com/resume>)

Contact

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Profile

Kevin has 15+ years of professional software engineering experience. He has a tenacious drive to learn and he has a passion for improving both the user and developer experience.

Personal Projects

Nessemble

<https://github.com/kevinsewyn/nessemble-rs>

A 6502 assembler for the Nintendo Entertainment System, written in Rust.

A ground-up rewrite of an original C assembler with a hand-written lexer, recursive-descent parser, and two-pass code generator supporting macros.

Ships a LSP for editor integration and a WebAssembly build that assembles ROMs entirely client-side in the browser.

Language(s):

Rust 6502 Assembly WASM TypeScript JavaScript
HTML CSS Bash

Tech:

Cargo Language Server Protocol WebAssembly Docker
GitHub Actions

pkmnapi

<https://github.com/pkmn-api/pkmnapi>

A Pokémon REST API that uses a Gen 1 Game Boy ROM as its data source.

Designed, documented, and implemented a RESTful API service.

Language(s):

Rust SQL

Tech:

Cargo PostgreSQL Heroku Github Actions

canvas-steg

<https://github.com/kevinsewyn/canvas-steg>

Execute and embed JavaScript hidden in steganographic images.

Proof-of-concept to demonstrate how unbelievably easy it is to embed JavaScript in an image and then execute it.

Language(s):

JavaScript

Tech:

Yarn NPM Webpack

Employment

Alloy 2023 - present

sr. software engineer ii

Kevin is a Senior Software Engineer on the GenAI squad, where he designs and builds the LLM and agentic platform behind Alloy's AI-assisted identity risk and fraud review products.

He champions the use of Server-Sent Events for LLM response streaming and contributes heavily to Alloy's in-house agent harness and tool-calling framework as well as an evaluations framework for automatic prompt optimization and data-driven model selection.

Language(s):

TypeScript JavaScript PostgreSQL HCL YAML Bash
HTML CSS SASS

Tech:

React NodeJS AWS Bedrock S3 SQS Textract
PostgreSQL Redis OpenSearch Snowflake Redshift
Kubernetes Helm Terraform Docker Datadog REST
Server-Sent Events Vite Storybook NPM Git
GitHub Actions Figma

Kevin regularly works with stakeholders on Product and Design teams to scope and deliver features through in-depth technical specifications and code reviews.

Capsule 2020 - 2023

staff software engineer

Kevin was a Staff Software Engineer/Tech Lead on the Internal Operations team.

He regularly worked with stakeholders on the Product and Design teams to ensure that new product features are delivered fast and right.

Kevin also led a squad of engineers to ensure solid and lasting architectural decisions are made through in-depth technical specifications and code reviews.

Language(s):

Python PostgreSQL Kotlin Bash JavaScript
TypeScript HTML CSS SASS

Tech:

Django AWS Cognito ElastiCache Elasticsearch RDS
SQS SNS CloudWatch DynamoDB S3 React
NodeJS NPM Webpack REST Git Docker Make
Jenkins CodeFresh Prometheus Grafana Loki
Terraform

Amper Music 2016 - 2020

sr. frontend engineer

Kevin worked as the sole frontend developer of Amper Music's enterprise AI music composition web app. Having inherited a legacy React codebase, 2 major versions of the application were released during his tenure including numerous minor versions.

He also worked closely with the Systems Architect/API Developer to test, document, and ultimately consume the company's public and internal web APIs.

Kevin utilized Rust to generate WebAssembly to create extremely performant mission-critical components.

Language(s):

JavaScript TypeScript HTML CSS SASS Python
Rust WASM Bash

Tech:

React NodeJS Yarn NPM Webpack Babel REST
GraphQL Git Vagrant Ubuntu Linux Cargo Make

Penguin Random House 2014 - 2016

↳ Fodors Travel, Living Language

sr. web developer

Kevin was the lead frontend web developer for Fodor's Travel and contributed greatly to Living Language.

He assisted in converting a legacy ColdFusion codebase to PHP/MySQL.

Kevin disassembled/reverse-engineered a series of Flash games and converted them to a custom JavaScript framework for maximum browser compatibility without breaking the existing API endpoints.

In addition to regular web development duties, Kevin also developed HTML emails and coordinated large, regularly scheduled email blasts.

Language(s):

JavaScript ActionScript Flash HTML CSS SASS
PHP ColdFusion SQL

Tech:

NodeJS Grunt Haxe MySQL SVN

TSL Marketing 2012 - 2014

sr. web developer

Kevin developed websites, blogs, landing pages, and HTML emails for clients in the technology sector.

He also performed SEO keyword research and optimization.

In addition to developing websites, he also trained clients on proper use, care, and customization of their website or blog through a CMS.

Not limited to digital, Kevin also aided in the fulfillment of dimensional mail campaigns (postage, shipping).

Language(s):

JavaScript HTML CSS PHP SQL

Tech:

Wordpress MySQL

Freelance 2010 - 2012

freelance web developer

Kevin created and maintained websites in a timely fashion and worked closely with clients to deliver great experiences for their users.

Language(s):

JavaScript PHP SQL

Tech:

Wordpress MySQL

Shenandoah Conservatory 2006 - 2010

bachelor of fine arts in musical theatre

Urbana High School 2002 - 2006

high school diploma

Language(s):

6502 Assembly Bash C CSS HCL HTML JavaScript Kotlin Lua PHP Python Rust SASS SQL TypeScript WASM YAML

Tech:

Adobe Creative Suite AWS Bedrock Cargo Claude Code Cursor Datadog Django Docker DynamoDB Figma Git GitHub Actions Helm Kubernetes Make MCP NestJS NodeJS NPM OpenSearch PostgreSQL React Redis REST S3 Server-Sent Events Snowflake SQS Storybook Terraform Vite WebAssembly

Kevin is passionate about music, The Office, The Legend of Zelda, and Corgis. He can grow a beard with great haste and loves singing and playing the guitar (poorly).

Education

Skills

Interests